

Mohammed Afaan Ansari

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ML Engineer | Generative Imaging & Computer Vision. Fine-tuned diffusion, GAN, and autoregressive models for denoising, super-resolution, in-painting; deployed to GPU/edge with >40% latency cut and >99% cost savings via mixed-precision, ONNX/TorchScript.

EDUCATION

University of Maryland, College Park

Aug 2024 – May 2026

M.S. Computer Science | GPA: 3.81/4.0 | Advisor: Prof. C. Buntain (CLIP/iSchool)

College Park, MD

Sardar Patel Institute of Technology, Mumbai University

Jul 2019 – Jun 2022

B.Tech. Computer Engineering | GPA: 8.9/10 | ACM/Springer publications: NLP & blockchain

Mumbai, India

TECHNICAL SKILLS

- **Generative Imaging & CV:** Python, PyTorch, TensorFlow, OpenCV, diffusers, HuggingFace; UNet/DDPM diffusion, GANs, autoregressive image models, CLIP, VLMs (InternVL, LLaVA, Pixtral); image restoration (denoising, super-resolution, in-painting); PSNR/SSIM evaluation.
- **Optimization & Deployment:** ONNX, TorchScript, mixed-precision, quantization, LoRA/PEFT; GPU profiling; mobile/edge deployment, vLLM; GCP/AWS, Docker, Kubernetes, CI/CD.

SELECTED PROJECTS & PUBLICATIONS

- **UNet Diffusion — Low-Light Image Restoration:** Conditioned DDPM trained on 3K-image smartphone dataset; **+3 dB PSNR, +0.02 SSIM** vs. baseline; mixed-precision ONNX inference <150ms mobile-GPU.
- **Efficient VLM Inference for Image-Grounded Generation & Retrieval** (preprint, 2025) — GPU-optimized inference, quality-latency tradeoffs, deployment-ready evaluation.
- **ChartDesign** (*COLM 2026, lead author*) — LoRA-tuned LLMs (CSV → chart JSON); 84% accuracy vs. 55% zero-shot.
- **Quantifying the Transience of Social Web Datasets** — ASONAM 2023 (ACM/IEEE), *2nd Best Paper*.

EXPERIENCE & RESEARCH

CSSI Inc.

Jun 2025 – Aug 2025

Data Science Intern

Washington, D.C.

- Architected anomaly-detection on 10M+ FAA aviation-weather records (90%+ recall); built FAA-compliant ETL + REST APIs powering compliance dashboards.

University of Maryland — CLIP Lab & iSchool

Aug 2024 – Present

Graduate Research Assistant — Multimodal ML (Prof. C. Buntain)

College Park, MD

- Optimized vision+language models in production (10K+ docs) via LoRA fine-tuning + ONNX acceleration; lifted output quality 25% and cut inference latency 40% — same patterns used for image-conditioned generation.

Georgia Tech & Mayo Clinic

Dec 2022 – Dec 2023

Research Assistant — Biomedical ML (Prof. Joy Arulraj)

Atlanta, GA (Remote)

- Built CNN-based MRI tumor-detection (85% accuracy on 1K patients) targeted for clinical integration at Mayo Clinic; benchmarked GPT-4 vs. ONNX-quantized vision encoders within EVADB (150ms → 90ms).

MSCI Inc.

Jul 2022 – Aug 2024

Data Science Technology Associate | Star Performer 2023, top of 1,200+

Mumbai, India

- Promoted to Associate in 1yr (vs. typical 3–4); built GCP pipeline processing 100K+ research reports (QA errors -30%); deployed ONNX-quantized inference (<100ms), cutting cost 99.98% per 1K items.

MSCI Sustainability Institute

Jul 2025 – Sep 2025

Research Participant — Green CapEx Initiative

Remote

- Computed ESG/CapEx metrics from 10K+ records adopted into MSCIT's Green CapEx taxonomy used by institutional investors.

Rochester Institute of Technology

Jul 2022 – Jun 2023

Research Assistant — Social Data Mining (Prof. A. KhudaBukhsh)

Rochester, NY (Remote)

- Co-authored ACM/IEEE paper on tweet lifespan using BERT; *2nd Best Paper, ASONAM 2023*.

LEADERSHIP & ACHIEVEMENTS

- Winner — Univ. Hackathon (LLM QnA), Pharma Hackathon; mentored 40+ ML/DS students (10+ at FAANG); raised \$50K (+70%) for NGO programs.